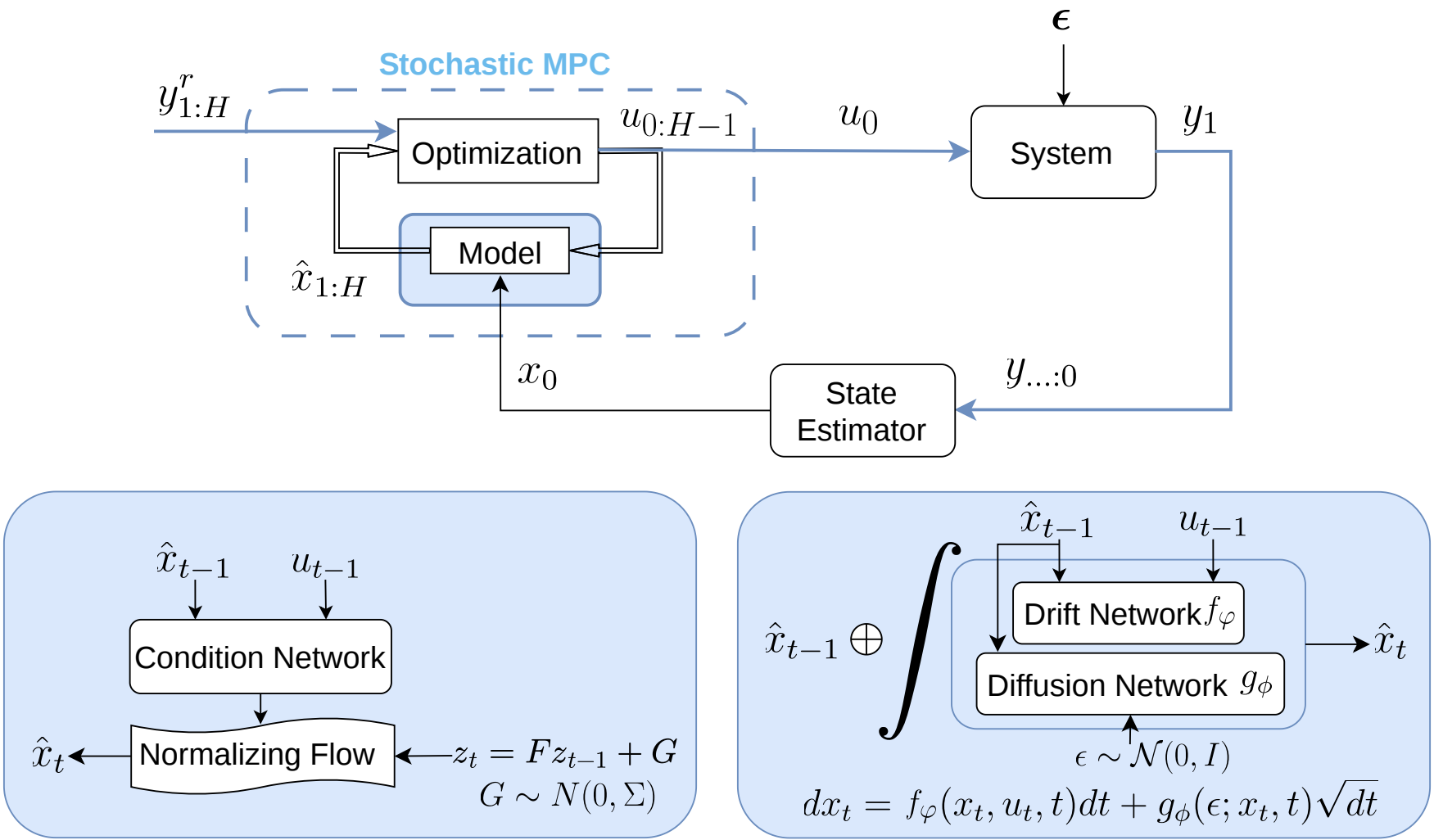
Stochastic MPC
$y^r_{1:H}$
Optimization
$u_{0:H-1}$
u_0
ϵ
System
y_1
Model
$\hat{x}_{1:H}$
x_0
State Estimator
$y_{...:0}$
$\hat{x}_{t-1}$
u_{t-1}
Condition Network
$\hat{x}_t$
Normalizing Flow
$z_t = Fz_{t-1} + G$
$G \sim N(0, \Sigma)$
$\hat{x}_{t-1}$
u_{t-1}
$\hat{x}_{t-1} \oplus \int$
Drift Network f_φ
Diffusion Network g_ϕ
$\hat{x}_t$
$\epsilon \sim \mathcal{N}(0, I)$
$dx_t = f_\varphi(x_t, u_t, t)dt + g_\phi(\epsilon; x_t, t)\sqrt{dt}$